

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Supplementary Examination – Summer 2022 Course: B. Tech. Branch :Mechanical Engineering Semester :VI Subject Code & Name: Mechanical Measurement BTMEC604D Max Marks: 60 Date: Duration: 3 Hr.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/CO)	Marks
Q. 1	Solve Any Two of the following.		
A)	Define 1)Hysteresis 2)Linearity 3) Resolution 4) Threshold 5) loading effect	CO1	6
B)	With the help of neat sketch explain general measurement system.	CO1	6
C)	Define error and explain different types of error.	CO1	6
Q.2	Solve Any Two of the following.		
A)	Enlist the different types of linear measurement instruments and with neat sketch explain Micrometer principle.	CO2	6
B)	How the wringing is possible in slip gauges?	CO2	6
C)	What is optical flat? How they are used for checking flatness?	CO2	6
Q. 3	Solve Any One of the following.		
A)	What are the different types of pressure measuring devices? Explain one	CO3	6
B)	Explain the concept of LVDT type pressure transducer.	CO3	6
C)	How Rota-meter is used to measure flow of fluid?	CO3	6
Q.4	Solve Any Two of the following.		
A)	How force is measured with the help of Load Cell?	CO3	6
B)	With help of neat sketch explain torsion bar dynamometer.	CO3	6
C)	How the angular speed is measured with electromagnetic transducers?	CO3	6
Q. 5	Solve Any One of the following.		
A)	How seismic accelerometer works?	CO3	6
B)	What are different temperature measuring devices? Explain thermocouple	CO3	6
C)	How temperature measuring devices are calibrated?	CO3	6
	*** End ***		

